CEMAT104 · DIFFERENTIAL EQUATIONS · AY 2026–2027, 1ST TERM

Sagot sa Bagong Reviewer ni Sam

Complete, step-by-step solutions to the **Long Test** (25 items) and the **Midterm Examination** (42 items), with simple explanations for every item.

How to use this reviewer

1. **Start with the Quick Review** (Part 0). It has every rule and method used in both exams on two pages.
2. **Try each item yourself first**, then check the solution. Every item has: Solution the full steps, Answer the final answer in a box, and In simple words the idea behind it.
3. **Night before the exam?** Read only the Quick Review and the **Answer Key at a Glance** at the end.

Long Test

A · Order & degree	1–6
B · Linear or nonlinear	7–12
C · Verifying solutions	13–17
D · Eliminating constants	18–21
E · Families of curves	22–25

Midterm Exam

A · DE terminologies	1–14
B · Order & degree	15–18
C · Verifying solutions	19–20
D · Eliminating constants	21–24
E · Families of curves	25–30
F · Solving DEs	31–42

Heads-up: typos in the test papers. A few items look mistyped. Each one is flagged where it appears and solved using the most likely intended version: LT #15 ($4' \to 4y'$), LT #19 (repeated cosine term), ME #25 (value of α missing, kept as α), ME #33 (second $dx \to dy$).

PART 0 Quick Review: everything you need on two pages

1 · Order and Degree

- **Order** = the highest derivative that appears.
- **Degree** = the power of that highest derivative, after the equation is written as a polynomial in the derivatives (no radicals, fractional or negative powers, and no derivative inside ln, sin, e^x, ...).
- To clear radicals and fractional powers, **raise both sides to a power**. To clear negative powers, **multiply through**.
- If a derivative sits inside ln, sin, etc., **isolate and invert** it, e.g. $\ln y'' = u \Rightarrow y'' = e^u$.

2 · Linear vs. Nonlinear (in *y*)

A DE is **linear** if it can be written as

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = f(x)$$

It is **nonlinear** if you see any of these:

- a power of y or a derivative other than 1: y^2, $(y')^3$, $\sqrt{y'}$
- a product such as $y\,y'$ or $y\,y''$
- a function of y: $\sin y$, e^y, $\cos(x+y)$, $\ln y$
- y in a denominator: $\dfrac{1}{x+y}$

Coefficients may depend on *x* in any way (e^x, x^2, $\sin x$ are all fine).

3 · Verifying a Solution

1. Differentiate the proposed solution as many times as needed.
2. Substitute into the DE.
3. Simplify. If LHS = RHS for all *x*, it **is** a solution.

If the solution is **implicit** ($F(x,y) = C$), differentiate it implicitly and show you get back the DE.

4 · Eliminating Arbitrary Constants

Rule: n arbitrary constants → differentiate n times → DE of order n.

Shortcut for exponentials: read the roots, then build the characteristic equation.

Solution form	Roots	DE
$Ae^{r_1x} + Be^{r_2x}$	r_1, r_2	$y'' - (r_1{+}r_2)y' + r_1r_2y = 0$
$(A + Bx)e^{rx}$	r, r	$y'' - 2ry' + r^2y = 0$
$e^{\alpha x}(A\cos\beta x + B\sin\beta x)$	$\alpha \pm \beta i$	$y'' - 2\alpha y' + (\alpha^2{+}\beta^2)y = 0$

Mixed forms (e.g. $c_1x^2 + c_2e^{kx}$): use the determinant (Wronskian) method, shown in LT #21.

5 · DE of a Family of Curves

1. Write the equation of the family. Count the **arbitrary** constants (the ones that change from curve to curve).
2. Fixed values (r, a, Q, k, α) are **not** eliminated. They stay in the answer.
3. Differentiate as many times as there are arbitrary constants.
4. Eliminate the arbitrary constants algebraically.

6 · Solving First-Order DEs: How to Recognize the Method

Method	How to recognize it	What to do
Separable	Can be arranged as $f(x)\,dx + g(y)\,dy = 0$	Integrate each part: $\int f\,dx + \int g\,dy = C$
Homogeneous	M and N are homogeneous of the same degree (the equation only depends on y/x)	Let $y = vx$, $dy = v\,dx + x\,dv$, and then separate
Exact	$M\,dx + N\,dy = 0$ with $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$	$F = \int M\,dx + g(y)$, find g from $F_y = N$; answer $F = C$
Linear	$y' + P(x)\,y = Q(x)$	IF $= e^{\int P\,dx}$; then $y \cdot \text{IF} = \int Q \cdot \text{IF}\,dx + C$
Bernoulli	$y' + P(x)\,y = Q(x)\,y^n$	Let $v = y^{1-n}$; the equation becomes linear in v

Initial condition (e.g. "when x = 0, y = 1"): substitute it into the general solution to get the value of C, which gives the particular solution.

PART 1 Long Test: Complete Solutions

A. Identify the order and degree (solve, if applicable)

1 $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$

SOLUTION

It has partial derivatives (u depends on both x and y), so it is a **partial** DE (PDE).

Highest derivative: second-order partials u_{xx}, u_{yy}, which appear to the first power.

✓ ANSWER

PDE · **Order 2, Degree 1** (this is Laplace's equation)

IN SIMPLE WORDS Two "∂^2" terms mean second derivatives, so the order is 2. They are not squared, so the degree is 1.

2 $\sin\left(\dfrac{dy}{dx}\right) = x$

SOLUTION

The derivative is *inside* a sine, so the degree can't be read directly. Take the inverse sine of both sides:

$$\frac{dy}{dx} = \sin^{-1} x$$

Now it is a polynomial in y' of power 1. It can also be **solved** by integrating (integration by parts):

$$y = \int \sin^{-1} x \, dx = x \sin^{-1} x + \sqrt{1 - x^2} + C$$

✓ ANSWER

Order 1, Degree 1 (after writing it as $y' = \sin^{-1} x$)

Solution: $y = x \sin^{-1} x + \sqrt{1 - x^2} + C$

IN SIMPLE WORDS You can't count the "power" of something trapped inside sin. Free it first with $\sin^{-1}$, then count.

3 $\dfrac{3x^2}{\sqrt{\dfrac{d^3y}{dx^3} + \left(\dfrac{d^2y}{dx^2}\right)^2}} = 7y$

SOLUTION

Clear the fraction, then square both sides to remove the radical:

$$3x^2 = 7y\sqrt{y''' + (y'')^2} \implies 9x^4 = 49y^2\left[\, y''' + (y'')^2 \,\right]$$

Highest derivative: y''', which appears to the power 1.

✓ ANSWER

Order 3, Degree 1

IN SIMPLE WORDS The $(y'')^2$ has a power of 2, but y'' is *not* the highest derivative. The degree only looks at the highest one, y'''.

4 $\ln\left(\frac{d^2y}{dx^2}\right) + \frac{dy}{dx} = x$

SOLUTION

Isolate the logarithm and take e of both sides:

$$\ln y'' = x - y' \implies y'' = e^{x-y'}$$

The highest derivative y'' now appears to the power 1.

✓ ANSWER

Order 2, Degree 1 (after writing it as $y'' = e^{x-y'}$)

IN SIMPLE WORDS Same trick as #2: a derivative inside ln is freed with $e^{(\,)}$.

5 $\left(\frac{d^2y}{dx^2}\right)^{-1} + \frac{dy}{dx} = x\frac{d^2y}{dx^2} + \cos x$

SOLUTION

A negative power is not allowed, so multiply every term by y'':

$$1 + y'\,y'' = x\,(y'')^2 + y''\cos x$$

Highest derivative y''; its highest power is now 2.

✓ ANSWER

Order 2, Degree 2

IN SIMPLE WORDS $(y'')^{-1}$ means $\frac{1}{y''}$. Multiplying by y'' clears it, but the term $x\,y''$ becomes $x(y'')^2$. That is where the degree 2 comes from.

6 $\left[y + x\left(\frac{d^2y}{dx^2}\right)^2\right]^{4/3} = x\left(\frac{dy}{dx}\right)$

SOLUTION

Remove the fractional power by cubing both sides:

$$\left[y + x(y'')^2\right]^4 = x^3(y')^3$$

Expanding the left side, the highest power of y'' is $\left[(y'')^2\right]^4 = (y'')^8$.

✓ ANSWER

Order 2, Degree 8

IN SIMPLE WORDS The power 4/3 goes away when you raise it to 3, which leaves a power of 4. Inside, y'' is already squared, so 2 × 4 = 8.

B. Linear or nonlinear? Give the reason for nonlinearity

#	DE	Answer	Reason
7	$y''' + e^x y' + y = \sin x$	**LINEAR**	y, y', y''' are all to the power 1 and not multiplied together. The coefficient e^x depends only on x, which is allowed.

8	$x\dfrac{d^2y}{dx^2} + y = 7$	**LINEAR**	Of the form $a_2(x)y'' + a_0(x)y = f(x)$ with $a_2 = x$, $a_0 = 1$, $f = 7$.
9	$\sqrt{\dfrac{dy}{dx}} + 3y = \cos x$	**NONLINEAR**	y' is raised to the power ½ (it is inside a square root).
10	$y'' + \cos(x + y) = e^x$	**NONLINEAR**	y appears inside a **cosine** (a transcendental function of y).
11	$\dfrac{dy}{dx} + \dfrac{3x^2 - 1}{x + y} = 0$	**NONLINEAR**	y is in a **denominator**. Clearing it gives $(x + y)y' + 3x^2 - 1 = 0$, which has the product $y\,y'$.
12	$(x^2 + y^2)\,dx + 2xy\,dy = 0$	**NONLINEAR**	As $y' = -\dfrac{x^2 + y^2}{2xy}$, it contains y^2 and the product $y\,y'$ (after clearing: $2xy\,y' + x^2 + y^2 = 0$).

IN SIMPLE WORDS Bonus for #12: it is **exact** ($M_y = 2y = N_x$), so it can be solved: $\dfrac{x^3}{3} + xy^2 = C$.

IN SIMPLE WORDS Quick test: look only at the terms that contain *y*. If each one is just (function of x) × (y or a derivative of y), the DE is linear.

C. Verify the solutions

13 Is $x^2 = 4 - 4y$ a particular solution of $\left(y + \sqrt{x^2 + y^2}\right)dx - x\,dy = 0$?

SOLUTION

Step 1: Express y and dy. $y = 1 - \dfrac{x^2}{4}, \qquad dy = -\dfrac{x}{2}\,dx$

Step 2: Simplify the radical.

$$x^2 + y^2 = x^2 + 1 - \frac{x^2}{2} + \frac{x^4}{16} = 1 + \frac{x^2}{2} + \frac{x^4}{16} = \left(1 + \frac{x^2}{4}\right)^2 \Rightarrow \sqrt{x^2 + y^2} = 1 + \frac{x^2}{4}$$

Step 3: Substitute into the DE as printed (with "+").

$$\left(1 - \tfrac{x^2}{4} + 1 + \tfrac{x^2}{4}\right)dx - x\left(-\tfrac{x}{2}\right)dx = \left(2 + \tfrac{x^2}{2}\right)dx \neq 0$$

Check with "−" instead: $y - \sqrt{x^2 + y^2} = -\dfrac{x^2}{2}$, so $-\dfrac{x^2}{2}dx + \dfrac{x^2}{2}dx = 0$ ✓

✓ ANSWER

NO. $x^2 = 4 - 4y$ does <u>not</u> satisfy $\left(y + \sqrt{x^2 + y^2}\right)dx - x\,dy = 0$, because the left side reduces to $\left(2 + \tfrac{x^2}{2}\right)dx \neq 0$.

It **is** the particular solution of $\left(y - \sqrt{x^2 + y^2}\right)dx - x\,dy = 0$ (see Midterm #39).

IN SIMPLE WORDS The trick is noticing that $x^2 + y^2$ is a perfect square here, so the radical disappears. Always substitute and simplify fully before you decide. A "not a solution" answer is valid if your algebra shows LHS ≠ 0.

14 Show that $2y = x^5 + cx^3$ is a solution of $(x^5 + 3y)\,dx - x\,dy = 0$.

SOLUTION

Write the DE as $x\,y' = x^5 + 3y$. From $2y = x^5 + cx^3$:

$$y = \frac{x^5 + cx^3}{2}, \qquad y' = \frac{5x^4 + 3cx^2}{2}$$

Left side: $x\,y' = \dfrac{5x^5 + 3cx^3}{2}$

Right side: $x^5 + 3y = x^5 + \dfrac{3x^5 + 3cx^3}{2} = \dfrac{5x^5 + 3cx^3}{2}$

✓ ANSWER

LHS = RHS $= \dfrac{5x^5 + 3cx^3}{2}$, so $2y = x^5 + cx^3$ **is a solution** ✓ (a general solution, since it has one arbitrary constant c).

IN SIMPLE WORDS Rewrite the dx/dy form as a y' form first; it makes the substitution cleaner.

15 Is $y = Ae^x + Bxe^x$ a solution of $y'' + 4y' - 4y = 0$?

Printed as "$y'' + 4' - 4y$". The missing y is assumed: $y'' + 4y' - 4y = 0$.

SOLUTION

$$y' = (A + B)e^x + Bxe^x, \qquad y'' = (A + 2B)e^x + Bxe^x$$

Substitute:

$$\begin{aligned} y'' + 4y' - 4y &= e^x\left[(A + 2B) + 4(A + B) - 4A\right] + xe^x\left[B + 4B - 4B\right] \\ &= (A + 6B)\,e^x + B\,xe^x \neq 0 \end{aligned}$$

✓ ANSWER

NO, it is not a solution (the result is not zero for arbitrary A, B).
$y = Ae^x + Bxe^x$ solves $y'' - 2y' + y = 0$ instead (double root $r = 1$).

IN SIMPLE WORDS Shortcut: $Ae^x + Bxe^x$ comes from a **double root r = 1**, i.e. $(r - 1)^2 = r^2 - 2r + 1$. The given DE has characteristic equation $r^2 + 4r - 4 = 0$, which is different, so the answer is NO right away.

16 Show that $2y = x^2 - 1 + 2e^{1-x^2}$ is a solution of $y' = x^3 - 2xy$.

SOLUTION

$$y = \frac{x^2 - 1}{2} + e^{1-x^2} \qquad \Longrightarrow \qquad y' = x - 2x\,e^{1-x^2}$$

Right side:

$$x^3 - 2x\left[\frac{x^2 - 1}{2} + e^{1-x^2}\right] = x^3 - x^3 + x - 2xe^{1-x^2} = x - 2xe^{1-x^2}$$

✓ ANSWER

LHS $= y' = x - 2xe^{1-x^2}$ = RHS, so **it is a solution** ✓ (a particular solution; there is no arbitrary constant).

IN SIMPLE WORDS Chain rule: $\dfrac{d}{dx}e^{1-x^2} = e^{1-x^2} \cdot (-2x)$.

17 Show that $(x - 3y + 2)\,dx + 3(x + 3y - 4)\,dy = 0$ has $\ln\left[(x - 1)^2 + 9(y - 1)^2\right] - 2\tan^{-1}\left[\dfrac{x - 1}{3(y - 1)}\right] = C$ as a solution.

SOLUTION

Step 1: Shift the variables to simplify: let $X = x - 1$, $Y = y - 1$ (so $dX = dx$, $dY = dy$).

$$x - 3y + 2 = X - 3Y, \qquad x + 3y - 4 = X + 3Y$$

So the DE becomes $(X-3Y)\,dX+3(X+3Y)\,dY=0$, and the solution becomes $\ln(X^2+9Y^2)-2\tan^{-1}\dfrac{X}{3Y}=C$.

Step 2: Differentiate the solution.

$$d\ln(X^2+9Y^2)=\frac{2X\,dX+18Y\,dY}{X^2+9Y^2}$$

$$d\tan^{-1}\frac{X}{3Y}=\frac{1}{1+\frac{X^2}{9Y^2}}\cdot\frac{3Y\,dX-3X\,dY}{9Y^2}=\frac{3Y\,dX-3X\,dY}{X^2+9Y^2}$$

Step 3: Combine (the differential of a constant is 0):

$$\frac{2X\,dX+18Y\,dY-2(3Y\,dX-3X\,dY)}{X^2+9Y^2}=0\implies(2X-6Y)\,dX+(6X+18Y)\,dY=0$$

Divide by 2: $(X-3Y)\,dX+3(X+3Y)\,dY=0$. Put back $X=x-1$, $Y=y-1$:

✓ ANSWER

$(x-3y+2)\,dx+3(x+3y-4)\,dy=0$, which is the original DE, so **it is a solution** ✓

IN SIMPLE WORDS For an implicit solution $F(x,y)=C$, take the differential of both sides; $dC=0$. If you get back the DE, it is verified. The shift $X=x-1$, $Y=y-1$ just makes the algebra shorter.

D. Eliminate the arbitrary constants

18 $y=Ae^{4x}+Bxe^{4x}$

SOLUTION

Two constants, so differentiate twice. A neat way: divide by e^{4x} first.

$$ye^{-4x}=A+Bx$$

Differentiate twice (a straight line has second derivative 0):

$$\frac{d}{dx}\left(ye^{-4x}\right)=(y'-4y)e^{-4x}=B$$

$$\frac{d}{dx}\left[(y'-4y)e^{-4x}\right]=(y''-4y'-4y'+16y)e^{-4x}=0$$

Check with the shortcut: double root $r=4$, so $(r-4)^2=r^2-8r+16$ ✓

✓ ANSWER

$y''-8y'+16y=0$

IN SIMPLE WORDS "e^{rx} and xe^{rx} together" always means a repeated root r. The DE is $y''-2ry'+r^2y=0$.

19 $x=c_1\cos\omega t+c_2\sin\omega t+c_3\cos\omega t$ (ω is a parameter)

The term $c_3\cos\omega t$ repeats $c_1\cos\omega t$. It is solved as printed.

SOLUTION

Combine like terms: $x=(c_1+c_3)\cos\omega t+c_2\sin\omega t=C_1\cos\omega t+c_2\sin\omega t$.

So there are only **two essential** arbitrary constants, and the DE is order 2.

$$\frac{dx}{dt}=-C_1\omega\sin\omega t+c_2\omega\cos\omega t$$

$$\frac{d^2x}{dt^2}=-\omega^2\left(C_1\cos\omega t+c_2\sin\omega t\right)=-\omega^2x$$

✓ ANSWER

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

IN SIMPLE WORDS c_1 and c_3 multiply the same function, so they act like one constant. The number of *essential* constants sets the order (2, not 3). ω is a parameter (fixed), so it stays in the answer.

20 $y = C_1 e^{3x}\cos 2x + C_2 e^{3x}\sin 2x$

SOLUTION

Differentiate once and write the result in terms of y:

$$y' = 3y + e^{3x}\left(-2C_1\sin 2x + 2C_2\cos 2x\right)$$

Let $z = y' - 3y = e^{3x}(-2C_1\sin 2x + 2C_2\cos 2x)$. Differentiate z:

$$z' = 3z + e^{3x}(-4C_1\cos 2x - 4C_2\sin 2x) = 3z - 4y$$

Substitute $z = y' - 3y$ and $z' = y'' - 3y'$:

$$y'' - 3y' = 3(y' - 3y) - 4y \implies y'' - 6y' + 13y = 0$$

Check with the shortcut: roots $3 \pm 2i$, so $r^2 - 2(3)r + (3^2 + 2^2) = r^2 - 6r + 13$ ✓

✓ ANSWER

$$y'' - 6y' + 13y = 0$$

IN SIMPLE WORDS $e^{\alpha x}(\cos\beta x,\ \sin\beta x)$ means roots $\alpha \pm \beta i$. The DE is $y'' - 2\alpha y' + (\alpha^2 + \beta^2)y = 0$. Here α = 3, β = 2.

21 $y = c_1 x^2 + c_2 e^{-2x}$

SOLUTION

The two functions are of different types, so use the **determinant (Wronskian) method**. Write y, y', y'' next to the derivatives of x^2 and e^{-2x}. The determinant must be zero:

$$\begin{vmatrix} y & x^2 & e^{-2x} \\ y' & 2x & -2e^{-2x} \\ y'' & 2 & 4e^{-2x} \end{vmatrix} = 0 \xrightarrow{\div\, e^{-2x}} \begin{vmatrix} y & x^2 & 1 \\ y' & 2x & -2 \\ y'' & 2 & 4 \end{vmatrix} = 0$$

Expand along the first column:

$$y(8x + 4) - y'(4x^2 - 2) + y''(-2x^2 - 2x) = 0$$

Divide by -2:

$$(x^2 + x)\,y'' + (2x^2 - 1)\,y' - (4x + 2)\,y = 0$$

Check $y = x^2$: $2(x^2 + x) + 2x(2x^2 - 1) - x^2(4x + 2) = 0$ ✓ **Check** $y = e^{-2x}$: $4(x^2 + x) - 2(2x^2 - 1) - (4x + 2) = 0$ ✓

✓ ANSWER

$$(x^2 + x)\,y'' + (2x^2 - 1)\,y' - (4x + 2)\,y = 0$$

IN SIMPLE WORDS Why the determinant works: y is a combination of x^2 and e^{-2x}, so the first column is a combination of the other two columns, which makes the determinant zero. This works for *any* $y = c_1 f(x) + c_2 g(x)$.

E. DE of families of curves (with graphs)

22 Straight lines at a fixed distance Q from the origin

SOLUTION

Family: a line with slope m and y-intercept b is $y = mx + b$. Its distance from the origin is

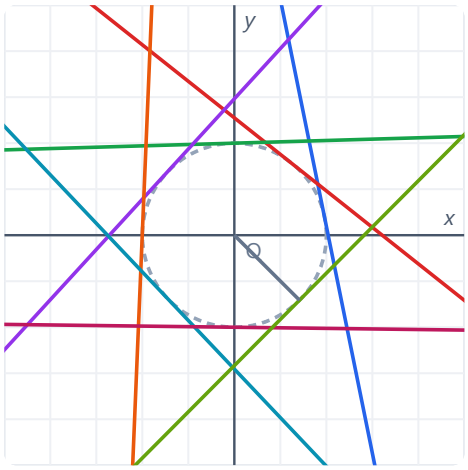
$$\frac{|b|}{\sqrt{1+m^2}} = Q \;\Rightarrow\; b = \pm Q\sqrt{1+m^2}$$

So $y = mx \pm Q\sqrt{1+m^2}$, with **one** arbitrary constant, m.

Eliminate m: for a line, $m = y'$. Substitute:

$$y = xy' \pm Q\sqrt{1+(y')^2}$$

Isolate the radical and square both sides:



All the lines touch the circle of radius Q.

✓ ANSWER

$$(y - xy')^2 = Q^2\left[1 + (y')^2\right]$$

IN SIMPLE WORDS Every line that stays exactly Q away from the origin is **tangent to the circle** $x^2 + y^2 = Q^2$. Q is fixed; only the slope changes from line to line.

23 Circles with fixed radius r, tangent to the y-axis

SOLUTION

If a circle touches the y-axis, its center is exactly r from it, at (r, k). Only k changes, so there is **one** arbitrary constant.

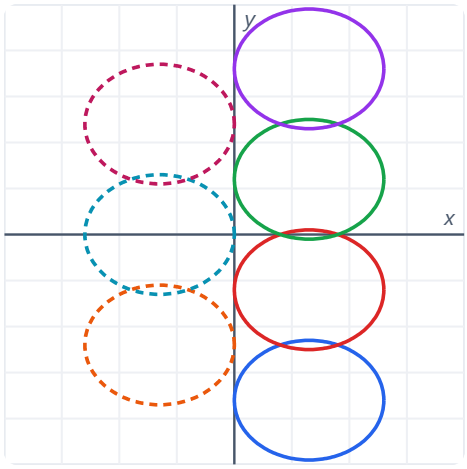
$$(x-r)^2 + (y-k)^2 = r^2$$

Differentiate: $2(x-r) + 2(y-k)y' = 0 \Rightarrow y - k = -\dfrac{x-r}{y'}$

Substitute back:

$$(x-r)^2 + \frac{(x-r)^2}{(y')^2} = r^2 \;\Rightarrow\; (x-r)^2\left[1+(y')^2\right] = r^2(y')^2$$

Expand $(x-r)^2$ and simplify:



Same radius; every circle touches the y-axis.

✓ ANSWER

$$(x-r)^2\left[1+(y')^2\right] = r^2(y')^2 \quad\Longleftrightarrow\quad (2rx - x^2)\left[1+(y')^2\right] = r^2$$

IN SIMPLE WORDS For circles on the left side of the axis (center $(-r, k)$), replace r with $-r$: $(x+r)^2[1 + (y')^2] = r^2(y')^2$.

24 Circles of radius unity (radius 1)

SOLUTION

The center (h, k) can be anywhere, so there are **two** arbitrary constants:

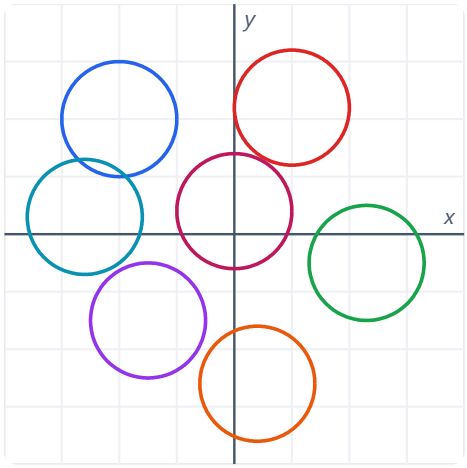
$$(x-h)^2 + (y-k)^2 = 1$$

1st derivative: $(x-h) + (y-k)y' = 0$

2nd derivative: $1 + (y')^2 + (y-k)y'' = 0 \Rightarrow y - k = -\dfrac{1+(y')^2}{y''}$

Then $x - h = -(y-k)y' = \dfrac{y'\left[1+(y')^2\right]}{y''}$. Substitute into the circle:

$$\frac{(y')^2\left[1+(y')^2\right]^2 + \left[1+(y')^2\right]^2}{(y'')^2} = 1$$



All circles have radius 1 and can be anywhere.

✓ ANSWER

$$\left[1+(y')^2\right]^3 = (y'')^2$$

IN SIMPLE WORDS This says the **curvature** $\dfrac{|y''|}{[1+(y')^2]^{3/2}}$ equals 1, which is exactly what "radius 1" means.

25 Parabolas with axis parallel to the x-axis, vertex-to-focus distance fixed as a

SOLUTION

The vertex (h, k) can be anywhere, so there are **two** arbitrary constants (a is fixed):

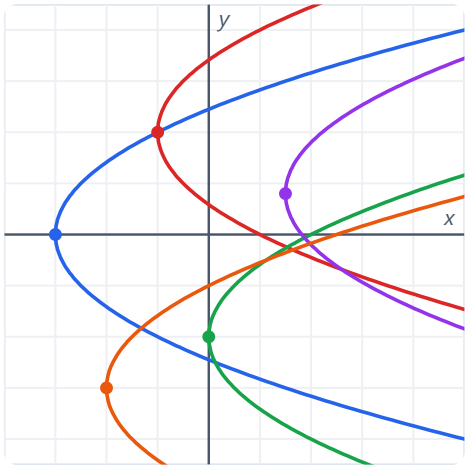
$$(y-k)^2 = 4a(x-h)$$

1st derivative: $2(y-k)y' = 4a \;\Rightarrow\; y - k = \dfrac{2a}{y'}$

2nd derivative: differentiate $y - k = 2a(y')^{-1}$:

$$y' = -\frac{2a\,y''}{(y')^2}$$

Multiply by $(y')^2$:



Same shape (same a), different vertices.

✓ ANSWER

$2a\,y'' + (y')^3 = 0$ (equivalently, with x as the dependent variable: $2a\dfrac{d^2x}{dy^2} = 1$)

IN SIMPLE WORDS Faster method: solve for x: $x = \dfrac{(y-k)^2}{4a} + h$. Differentiate twice with respect to y and both constants disappear: $\dfrac{d^2x}{dy^2} = \dfrac{1}{2a}$.

PART 2 Midterm Examination: Complete Solutions

A. Compare and contrast, with examples (1–14)

1–2 Dependent vs. Independent Variables

Dependent variable	Independent variable
The **unknown function** being differentiated; its value depends on the other variable.	The variable you differentiate **with respect to**; it varies freely.
It appears "on top" of the derivative: dy in $\frac{dy}{dx}$.	It appears "at the bottom": dx in $\frac{dy}{dx}$.

Example: $\frac{dy}{dx} = 3x^2$: y is dependent, x is independent. In $L\frac{di}{dt} + Ri = E$: current i is dependent, time t is independent.

Similarity: both are variables of the same DE, related through the derivative.

3–4 Ordinary vs. Partial DE

Ordinary DE (ODE)	Partial DE (PDE)
Only **one** independent variable.	**Two or more** independent variables.
Uses ordinary derivatives $\frac{dy}{dx}$.	Uses partial derivatives $\frac{\partial u}{\partial x}$.

Examples: ODE: $y'' + 4y = \sin x$. PDE: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (Laplace), $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ (heat equation).

Similarity: both relate an unknown function to its derivatives, and both are classified by order and degree.

5–6 Order vs. Degree of a DE

Order	Degree
The order of the **highest derivative** in the DE.	The **power** of that highest-order derivative, once the DE is a polynomial in its derivatives.
Always defined.	Not defined when a derivative is stuck inside a function (e.g. $\sin y'$) that can't be removed.

Example: $(y'')^3 + (y')^5 + y = x$ has **order 2** (highest is y'') and **degree 3** (power of y''). The 5 on y' doesn't matter.

Similarity: both are read from the highest derivative, and both are positive whole numbers.

7–8 General vs. Particular Solution

General solution	Particular solution
Contains **arbitrary constants** (as many as the order).	Has **no** arbitrary constants; they have been given specific values.
Represents a whole **family** of curves.	Represents **one** curve from that family.
Obtained by solving the DE.	Obtained from the general solution using **initial or boundary conditions**.

Example: $y' = 2x$ has general solution $y = x^2 + C$. With $y(0) = 3$: particular solution $y = x^2 + 3$.

Similarity: both satisfy the DE. The particular solution is a member of the general solution.

9–10 Arbitrary Constant, Particular Constant, and Parameter

Arbitrary constant	Particular constant	Parameter
An unspecified constant (C, c_1, A) in the general solution. It can take **any** value and is eliminated when forming a DE.	The **specific numerical value** an arbitrary constant takes once conditions are applied.	A constant that is **fixed by the problem** (a physical or geometric quantity). It is **not** eliminated and stays in the DE.
$y = Ce^{2x}$: C	$y(0) = 5 \Rightarrow C = 5$	ω in $x'' + \omega^2 x = 0$; R, L, E in a circuit; r in "circles of radius r"

Similarity: all three are constants (they don't change with x). The difference is **whether** they are free, fixed by conditions, or fixed by the problem.

11–12 Linear vs. Nonlinear DE

Linear	Nonlinear
Of the form $a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = f(x)$: y and its derivatives are to power 1, never multiplied together, and never inside functions.	Anything that is not linear: powers (y^2, $(y')^3$), products (yy'), or functions of y ($\sin y$, e^y).
Superposition works: sums of solutions are solutions (homogeneous case).	Superposition generally fails.

Examples: Linear: $y'' + x^2y' + e^x y = \cos x$. Nonlinear: $y'' + \sin y = 0$ (pendulum), $y' = y^2$.

Similarity: both can be of any order. "Linear" refers only to how y appears; the x-parts can be anything.

13–14 Homogeneous vs. Non-homogeneous DE

Homogeneous	Non-homogeneous
(Linear DE sense) The right side is zero: $a_n y^{(n)} + \cdots + a_0 y = 0$.	The right side is a nonzero $f(x)$: $a_n y^{(n)} + \cdots + a_0 y = f(x)$.
(1st-order sense) $M\,dx + N\,dy = 0$ where M, N are homogeneous functions of the **same degree**, e.g. $(x^2 + y^2)dx + 2xy\,dy = 0$ (degree 2). Solved with $y = vx$.	M and N are not homogeneous of the same degree, e.g. $(x + y + 1)dx + (x - y)dy = 0$.

Examples: $y'' - 4y = 0$ (homogeneous) vs. $y'' - 4y = e^x$ (non-homogeneous).

Similarity: the general solution of a non-homogeneous linear DE is (solution of the homogeneous part) + (one particular solution).

B. Identify the order and degree (solve, if applicable)

15 $\left(\frac{d^3y}{dx^3}\right)^{2/3} = \sqrt{1 + \frac{d^2y}{dx^2}}$

SOLUTION

The powers are $\frac{2}{3}$ and $\frac{1}{2}$. The LCD of 3 and 2 is 6, so raise both sides to the 6th power:

$$(y''')^4 = (1 + y'')^3$$

✓ ANSWER

Order 3, Degree 4

IN SIMPLE WORDS To remove all fractional powers at once, raise both sides to the LCD of the denominators. Here $\frac{2}{3} \times 6 = 4$.

16 $\left(\frac{d^2y}{dx^2}\right)^{-2/3} + \left(\frac{dy}{dx}\right)^{2/3} = x\left(\frac{d^2y}{dx^2}\right)^{1/3}$

SOLUTION

Let $p = y'$, $q = y''$. Multiply every term by $q^{2/3}$ to clear the negative power:

$$1 + p^{2/3}q^{2/3} = x\,q \implies (pq)^{2/3} = xq - 1$$

Cube both sides:

$$(y')^2(y'')^2 = (x\,y'' - 1)^3$$

Expanding the right side, the highest power of y'' is $x^3(y'')^3$.

✓ ANSWER

Order 2, Degree 3

IN SIMPLE WORDS Clear the negative power first (multiply), then the fractional power (cube). Then look for the highest power of the highest derivative on either side.

17 $\ln\left(\frac{d^2y}{dx^2}\right) + \frac{dy}{dx} = x$

SOLUTION

This is the same as Long Test #4: $\ln y'' = x - y' \Rightarrow y'' = e^{x-y'}$.

✓ ANSWER
Order 2, Degree 1

18 $\Delta^2\phi = 0$

SOLUTION

ϕ depends on several space variables (x, y, z), so this is a **PDE**.

As intended in class (Laplace's equation): $\nabla^2\phi = \dfrac{\partial^2\phi}{\partial x^2} + \dfrac{\partial^2\phi}{\partial y^2} + \dfrac{\partial^2\phi}{\partial z^2} = 0$, which is order 2, degree 1.

Strict notation: Δ alone is the Laplacian, so $\Delta^2 = \Delta\Delta$ is the *biharmonic* operator: $\dfrac{\partial^4\phi}{\partial x^4} + 2\dfrac{\partial^4\phi}{\partial x^2\partial y^2} + \dfrac{\partial^4\phi}{\partial y^4} = 0$ (in 2D), which is order 4, degree 1.

✓ ANSWER
PDE · **Order 2, Degree 1** (Laplace's equation, $\nabla^2\phi = 0$)
If Δ^2 is read strictly as the biharmonic operator: Order 4, Degree 1.

IN SIMPLE WORDS In most engineering classes $\Delta^2\phi$ or $\nabla^2\phi$ is used for Laplace's equation. Either way the degree is 1, because no derivative is raised to a power.

C. Verify the solutions

19 Show that $u(x, y) = \cos x \sinh y + \sin x \cosh y$ satisfies $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$.

SOLUTION

With respect to x (treat y as a constant):

$$u_x = -\sin x \sinh y + \cos x \cosh y, \qquad u_{xx} = -\cos x \sinh y - \sin x \cosh y = -u$$

With respect to y (treat x as a constant; $(\sinh y)' = \cosh y$, $(\cosh y)' = \sinh y$):

$$u_y = \cos x \cosh y + \sin x \sinh y, \qquad u_{yy} = \cos x \sinh y + \sin x \cosh y = u$$

✓ ANSWER
$u_{xx} + u_{yy} = -u + u = 0$ ✓, so u **is a solution** of Laplace's equation.

IN SIMPLE WORDS Differentiating sin/cos twice flips the sign; sinh/cosh twice does not. That's why the two parts cancel.

20 Verify that $J_0(x) = \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(n!)^2\, 2^{2n}}$ is a solution of $xy'' + y' + xy = 0$.

SOLUTION

First few terms: $J_0 = 1 - \dfrac{x^2}{4} + \dfrac{x^4}{64} - \dfrac{x^6}{2304} + \cdots$

Differentiate term by term (the $n = 0$ term is a constant, so it drops out):

$$y' = \sum_{n=1}^{\infty} \frac{(-1)^n\, 2n\, x^{2n-1}}{(n!)^2\, 2^{2n}}, \qquad y'' = \sum_{n=1}^{\infty} \frac{(-1)^n\, 2n(2n-1)\, x^{2n-2}}{(n!)^2\, 2^{2n}}$$

Combine $xy'' + y'$ (both now have x^{2n-1}):

$$xy'' + y' = \sum_{n=1}^{\infty} \frac{(-1)^n \left[2n(2n-1) + 2n\right] x^{2n-1}}{(n!)^2\, 2^{2n}} = \sum_{n=1}^{\infty} \frac{(-1)^n\, 4n^2\, x^{2n-1}}{(n!)^2\, 2^{2n}}$$

Simplify using $\dfrac{4n^2}{(n!)^2\, 2^{2n}} = \dfrac{1}{\left[(n-1)!\right]^2\, 2^{2(n-1)}}$, then shift the index with $m = n - 1$:

$$xy'' + y' = \sum_{m=0}^{\infty} \frac{(-1)^{m+1}\, x^{2m+1}}{(m!)^2\, 2^{2m}}$$

The last term: $xy = \displaystyle\sum_{m=0}^{\infty} \frac{(-1)^m\, x^{2m+1}}{(m!)^2\, 2^{2m}}$

The two series have the same terms with opposite signs ($(-1)^{m+1} = -(-1)^m$).

Quick check with the first terms: $y' = -\frac{x}{2} + \frac{x^3}{16} - \cdots$, $xy'' = -\frac{x}{2} + \frac{3x^3}{16} - \cdots$, $xy = x - \frac{x^3}{4} + \cdots$. Sum: $(-\frac{1}{2} - \frac{1}{2} + 1)x + (\frac{1}{16} + \frac{3}{16} - \frac{1}{4})x^3 = 0$ ✓

✓ ANSWER

$xy'' + y' + xy = 0$ ✓, so $J_0(x)$ **is a solution** (Bessel's equation of order zero).

IN SIMPLE WORDS With series, the goal is to make every sum use the same power of x (here x^{2m+1}) and the same starting index. Then compare the coefficients.

D. Eliminate the arbitrary constants

21 $y = Ae^{2x} + Bxe^{2x}$

SOLUTION

$ye^{-2x} = A + Bx$. Differentiate twice:

$$(y' - 2y)e^{-2x} = B \;\Rightarrow\; (y'' - 2y' - 2y' + 4y)e^{-2x} = 0$$

Shortcut check: double root $r = 2$, so $(r-2)^2 = r^2 - 4r + 4$ ✓

✓ ANSWER

$y'' - 4y' + 4y = 0$

22 $x = c_1 \cos \omega t + c_2 \sin \omega t$ (ω is a parameter)

SOLUTION

$$\dot{x} = -c_1\omega \sin \omega t + c_2\omega \cos \omega t, \qquad \ddot{x} = -\omega^2 \left(c_1 \cos \omega t + c_2 \sin \omega t\right) = -\omega^2 x$$

✓ ANSWER

$\dfrac{d^2x}{dt^2} + \omega^2 x = 0$

IN SIMPLE WORDS This is the **simple harmonic motion** equation (a spring–mass system).

23 $y = C_1 e^{2x} \cos 3x + C_2 e^{2x} \sin 3x$

SOLUTION

Same steps as Long Test #20. Let $z = y' - 2y = e^{2x}(-3C_1 \sin 3x + 3C_2 \cos 3x)$. Then

$$z' = 2z - 9y \;\Rightarrow\; y'' - 2y' = 2(y' - 2y) - 9y$$

Shortcut check: roots $2 \pm 3i$, so $r^2 - 4r + (4+9) = r^2 - 4r + 13$ ✓

✓ ANSWER
$y'' - 4y' + 13y = 0$

24 $y = c_1 x^2 + c_2 e^{2x}$

SOLUTION

$$\begin{vmatrix} y & x^2 & e^{2x} \\ y' & 2x & 2e^{2x} \\ y'' & 2 & 4e^{2x} \end{vmatrix} = 0 \xrightarrow{\div e^{2x}} \begin{vmatrix} y & x^2 & 1 \\ y' & 2x & 2 \\ y'' & 2 & 4 \end{vmatrix} = 0$$

Expand along the first column:

$$y(8x - 4) - y'(4x^2 - 2) + y''(2x^2 - 2x) = 0$$

Divide by 2:

$$(x^2 - x)\,y'' - (2x^2 - 1)\,y' + (4x - 2)\,y = 0$$

Check $y = x^2$: $2(x^2 - x) - 2x(2x^2 - 1) + x^2(4x - 2) = 0$ ✓ **Check** $y = e^{2x}$: $4(x^2 - x) - 2(2x^2 - 1) + (4x - 2) = 0$ ✓

✓ ANSWER
$(x^2 - x)\,y'' - (2x^2 - 1)\,y' + (4x - 2)\,y = 0$

IN SIMPLE WORDS Compare with Long Test #21: changing e^{-2x} to e^{2x} changes the signs in the determinant.

E. DE of families of curves (with graphs)

25 Straight lines with algebraic sum of intercepts fixed as α

The value after "fixed as" is blank on the paper. It is kept as the constant α (per the problem's wording).

SOLUTION

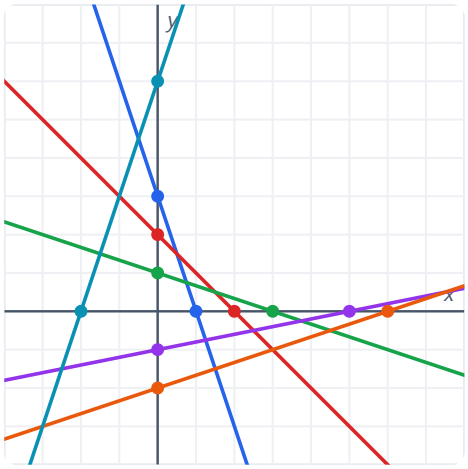
Intercepts a and b with $a + b = \alpha$, so $b = \alpha - a$ and there is **one** arbitrary constant, a:

$$\frac{x}{a} + \frac{y}{\alpha - a} = 1$$

Differentiate: $\dfrac{1}{a} + \dfrac{y'}{\alpha - a} = 0 \Rightarrow \alpha - a = -a\,y' \Rightarrow a = \dfrac{\alpha}{1 - y'}$

The y-intercept is $b = \alpha - a = \dfrac{-\alpha y'}{1 - y'} = \dfrac{\alpha y'}{y' - 1}$. A line is $y = (\text{slope})x + b$:

$$y = xy' + \frac{\alpha\,y'}{y' - 1}$$



α = 4: x-intercept + y-intercept = 4 for every line.

✓ ANSWER
$y = xy' + \dfrac{\alpha\,y'}{y' - 1} \quad \Longleftrightarrow \quad (y - xy')(y' - 1) = \alpha\,y'$

IN SIMPLE WORDS Check with α = 3, a = 1: the line $x + \frac{y}{2} = 1$, i.e. $y = 2 - 2x$, has $y' = -2$. RHS $= -2x + \frac{3(-2)}{-3} = -2x + 2$ ✓

26 All circles

SOLUTION

Center (h, k) and radius r are all free, so there are **three** arbitrary constants:

$$(x-h)^2 + (y-k)^2 = r^2$$

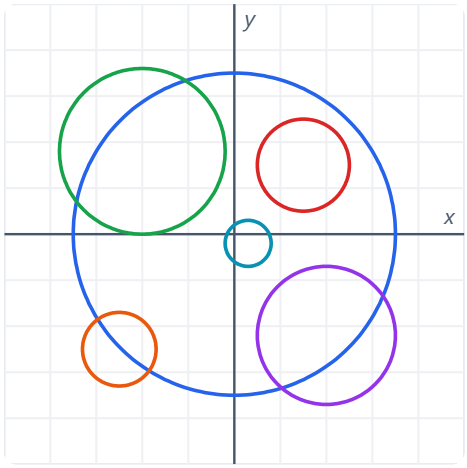
1st: $(x-h) + (y-k)y' = 0$

2nd: $1 + (y')^2 + (y-k)y'' = 0 \;\Rightarrow\; y - k = -\dfrac{1+(y')^2}{y''}$

3rd: $2y'y'' + y'y'' + (y-k)y''' = 0 \;\Rightarrow\; 3y'y'' + (y-k)y''' = 0$

Substitute $y - k$ from the 2nd derivative:

$$3y'y'' - \frac{\left[1+(y')^2\right]y'''}{y''} = 0$$



Any center, any radius.

✓ ANSWER

$$\left[1+(y')^2\right]y''' - 3y'(y'')^2 = 0$$

IN SIMPLE WORDS Three constants mean the 3rd derivative is needed. h and r disappear by the 3rd differentiation; the only one left to substitute is k.

27 All parabolas

SOLUTION

A parabola of any position and orientation is $(ax+by)^2 + cx + dy + e = 0$. After scaling, it has **four** arbitrary constants, so expect order 4.

Let $u = ax + by$ and $w = a + by'$ (so $u' = w$).

1st: $2uw + c + dy' = 0 \;\Rightarrow\; w(2bu + d) = ad - bc \equiv K$ (a constant)

2nd: $2w^2 + (2bu+d)y'' = 0 \;\Rightarrow\; y'' = -\dfrac{2w^2}{2bu+d} = -\dfrac{2w^3}{K}$

So $(y'')^{1/3}$ is a linear function of y': $(y'')^{1/3} = \lambda(a + by')$.

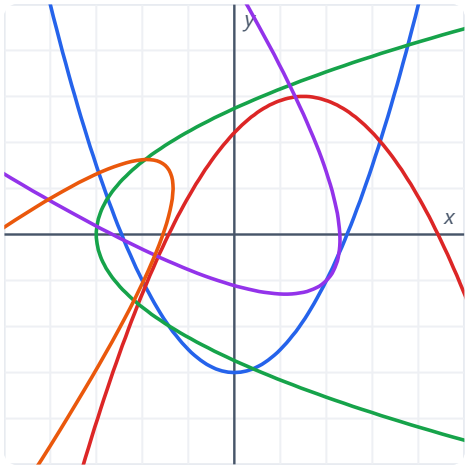
Differentiate:

$$\tfrac{1}{3}(y'')^{-2/3}y''' = \lambda b\, y'' \;\Rightarrow\; (y'')^{-5/3}y''' = 3\lambda b = \text{const.}$$

Differentiate once more to remove the last constant:

$$-\tfrac{5}{3}(y'')^{-8/3}(y''')^2 + (y'')^{-5/3}y^{(4)} = 0$$

Multiply by $3(y'')^{8/3}$:



Any vertex, any width, any direction.

✓ ANSWER

$$3\,y''\,y^{(4)} - 5\,(y''')^2 = 0 \qquad \left(\text{equivalently } \frac{d^2}{dx^2}\left[(y'')^{-2/3}\right] = 0\right)$$

IN SIMPLE WORDS Check with the sideways parabola $y = \sqrt{x}$: $y'' = -\frac{1}{4}x^{-3/2}$, $y''' = \frac{3}{8}x^{-5/2}$, $y^{(4)} = -\frac{15}{16}x^{-7/2}$. Then $3y''y^{(4)} = \frac{45}{64}x^{-5} = 5(y''')^2$ ✓. For an upright parabola, y'' is constant and every term is 0 ✓.

28 Parabolas with axis parallel to the y-axis, vertex-to-focus distance fixed as k

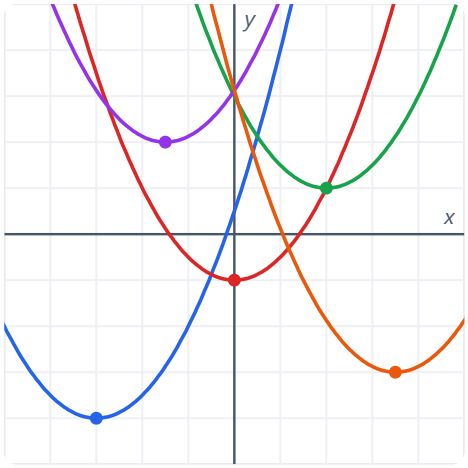
SOLUTION

The vertex (h, v) can be anywhere, so there are **two** arbitrary constants (k fixed):

$$(x-h)^2 = 4k(y-v)$$

1st: $2(x-h) = 4k\,y' \;\Rightarrow\; x-h = 2k\,y'$

2nd: $1 = 2k\,y''$



Same shape (same k), vertices anywhere.

✓ ANSWER

$2k\,y'' = 1 \quad \left(\text{i.e. } y'' = \dfrac{1}{2k}\right)$

IN SIMPLE WORDS This is the upright version of Long Test #25. Every parabola with the same k has the same constant "bend" y''.

29 Straight lines passing through a fixed point (a, b) in Quadrant III

SOLUTION

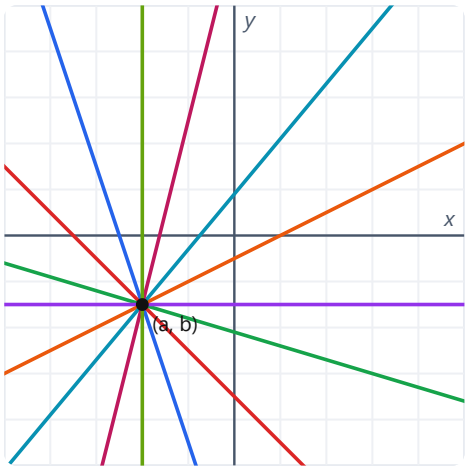
The point is fixed; only the slope m varies, so there is **one** arbitrary constant:

$$y - b = m(x-a)$$

Differentiate: $y' = m$. Substitute back:

$$y - b = y'(x-a)$$

Quadrant III means $a < 0$ and $b < 0$. Writing $a = -|a|$, $b = -|b|$: $(x + |a|)\,y' = y + |b|$.



A "star" of lines through one point in QIII.

✓ ANSWER

$(x-a)\,y' - (y-b) = 0 \quad \text{or} \quad (x-a)\,dy - (y-b)\,dx = 0, \qquad a < 0,\ b < 0$

IN SIMPLE WORDS a and b are given (fixed), so they stay in the DE. Only m is eliminated.

30 Parabolas with axis parallel to the x-axis

SOLUTION

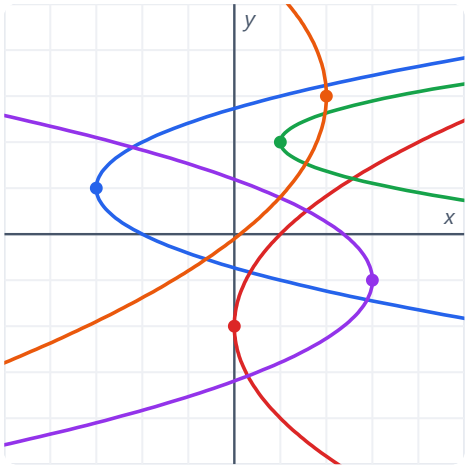
Now the width is free too, so there are **three** arbitrary constants. The easiest form solves for x:

$$x = Ay^2 + By + C \quad \Rightarrow \quad \frac{d^3x}{dy^3} = 0$$

To write this with y as the dependent variable, use $\dfrac{dx}{dy} = \dfrac{1}{y'}$ and differentiate again:

$$\frac{d^2x}{dy^2} = -\frac{y''}{(y')^3}, \qquad \frac{d^3x}{dy^3} = \frac{3(y'')^2 - y'y'''}{(y')^5}$$

Set it to zero:



Opening left or right; any vertex, any width.

✓ ANSWER

$$\frac{d^3x}{dy^3} = 0 \quad \Longleftrightarrow \quad y'\,y''' - 3\,(y'')^2 = 0$$

IN SIMPLE WORDS Another way: in LT #25, $2a\,y'' + (y')^3 = 0$ means $\dfrac{(y')^3}{y''} = -2a$ is constant. If a is free too, differentiate that once more and you get the same answer.

F. Solve the following DEs

31 $\sin x \sin y\,dx + \cos x \cos y\,dy = 0$

Separable

SOLUTION

Divide by $\cos x \sin y$ to separate:

$$\frac{\sin x}{\cos x}dx + \frac{\cos y}{\sin y}dy = 0 \;\Rightarrow\; \int \tan x\,dx + \int \cot y\,dy = 0$$

$$-\ln|\cos x| + \ln|\sin y| = \ln C \;\Rightarrow\; \ln\left|\frac{\sin y}{\cos x}\right| = \ln C$$

✓ ANSWER

$$\sin y = C\cos x$$

IN SIMPLE WORDS Remember: $\int \tan x\,dx = -\ln|\cos x|$ and $\int \cot y\,dy = \ln|\sin y|$. Writing the constant as $\ln C$ lets the logs combine neatly.

32 $y \ln x \ln y\,dx + dy = 0$

Separable

SOLUTION

Divide by $y \ln y$:

$$\ln x\,dx + \frac{dy}{y\ln y} = 0$$

Integrate. For the left, use parts: $\int \ln x\,dx = x\ln x - x$. For the right, let $u = \ln y$, $du = dy/y$: $\int \frac{du}{u} = \ln|\ln y|$.

$$x\ln x - x + \ln|\ln y| = C$$

✓ ANSWER

$x \ln x - x + \ln(\ln y) = C \qquad \text{or} \qquad \ln y = K\, e^{x} x^{-x}$

IN SIMPLE WORDS Check: differentiating $\ln(\ln y) = K' + x - x \ln x$ gives $\frac{y'}{y \ln y} = -\ln x$, i.e. $y' = -y \ln x \ln y$ ✓

33 $(x - y \ln y + y \ln x)\, dx + x(\ln y - \ln x)\, dy = 0$ — Homogeneous

Printed with "dx" twice. With two dx's it isn't a DE (there's no dy), so the second is taken as dy.

SOLUTION

Use $\ln y - \ln x = \ln \frac{y}{x}$: $\left(x - y \ln \frac{y}{x}\right) dx + x \ln \frac{y}{x}\, dy = 0$, which depends only on $\frac{y}{x}$, so it is **homogeneous**.

Let $y = vx$, $dy = v\, dx + x\, dv$:

$$(x - vx \ln v)\, dx + x \ln v\, (v\, dx + x\, dv) = 0$$

$$x\, dx \underbrace{- vx \ln v\, dx + vx \ln v\, dx}_{=0} + x^2 \ln v\, dv = 0 \;\Rightarrow\; \frac{dx}{x} + \ln v\, dv = 0$$

Integrate ($\int \ln v\, dv = v \ln v - v$):

$$\ln x + v \ln v - v = C$$

Put back $v = y/x$ and multiply by x:

✓ ANSWER

$x \ln x + y \ln \frac{y}{x} - y = Cx$

IN SIMPLE WORDS In homogeneous problems, the $v\, dx$ terms often cancel perfectly. That's the sign you're on the right track.

34 $y' = x - 2y \cot 2x$ — Linear

SOLUTION

Standard form: $y' + (2 \cot 2x)\, y = x$, so $P = 2 \cot 2x$, $Q = x$.

Integrating factor: $\text{IF} = e^{\int 2 \cot 2x\, dx} = e^{\ln|\sin 2x|} = \sin 2x$

$$\frac{d}{dx}(y \sin 2x) = x \sin 2x$$

Integrate by parts ($u = x$, $dv = \sin 2x\, dx$):

$$y \sin 2x = \int x \sin 2x\, dx = -\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x + C$$

✓ ANSWER

$y \sin 2x = \frac{1}{4} \sin 2x - \frac{x}{2} \cos 2x + C \qquad \text{or} \qquad y = \frac{1}{4} - \frac{x}{2} \cot 2x + C \csc 2x$

IN SIMPLE WORDS $\int \cot 2x\, dx = \frac{1}{2} \ln|\sin 2x|$, so $\int 2 \cot 2x\, dx = \ln|\sin 2x|$ and IF $= \sin 2x$.

35 $(\cos x \cos y - \cot x)\, dx - \sin x \sin y\, dy = 0$ — Exact

SOLUTION

$M = \cos x \cos y - \cot x$, $N = -\sin x \sin y$

Test: $\frac{\partial M}{\partial y} = -\cos x \sin y = \frac{\partial N}{\partial x}$ ✓ Exact.

Integrate M with respect to x:

$$F = \int (\cos x \cos y - \cot x)\, dx = \sin x \cos y - \ln|\sin x| + g(y)$$

Find g: $F_y = -\sin x \sin y + g'(y) = N = -\sin x \sin y \;\Rightarrow\; g'(y) = 0$

✓ ANSWER

$\sin x \cos y - \ln|\sin x| = C \qquad \text{or} \qquad \sin x \cos y = \ln(C \sin x)$

36 $2x^3y' = y(y^2 + 3x^2)$ — Bernoulli

SOLUTION

Divide by $2x^3$ and rearrange:

$$y' - \frac{3}{2x}\, y = \frac{1}{2x^3}\, y^3 \qquad (n = 3)$$

Let $v = y^{1-3} = y^{-2}$, so $v' = -2y^{-3}y'$. Multiply the DE by $-2y^{-3}$:

$$v' + \frac{3}{x}\, v = -\frac{1}{x^3}$$

This is linear in v. IF $= e^{\int 3/x\, dx} = x^3$:

$$(x^3 v)' = -1 \;\Rightarrow\; x^3 v = -x + C \;\Rightarrow\; \frac{x^3}{y^2} = C - x$$

✓ ANSWER

$y^2 = \dfrac{x^3}{C - x} \qquad \text{or} \qquad x^3 = y^2(C - x)$

IN SIMPLE WORDS You can also solve this as a homogeneous equation ($y = vx$) and get the same result. Bernoulli is quicker: the y^3 on the right is the clue.

37 $(2a^2 - r^2)\, dr = r^3 \sin\theta\, d\theta$; when $\theta = 0$, $r = a$ — Separable + IC

SOLUTION

Divide by r^3:

$$\left(\frac{2a^2}{r^3} - \frac{1}{r}\right) dr = \sin\theta\, d\theta$$

Integrate:

$$-\frac{a^2}{r^2} - \ln r = -\cos\theta + C$$

Apply $\theta = 0$, $r = a$: $-1 - \ln a = -1 + C \;\Rightarrow\; C = -\ln a$

$$-\frac{a^2}{r^2} - \ln r = -\cos\theta - \ln a \;\Rightarrow\; \frac{a^2}{r^2} + \ln r - \ln a = \cos\theta$$

✓ ANSWER

$\dfrac{a^2}{r^2} + \ln\dfrac{r}{a} = \cos\theta$

IN SIMPLE WORDS Check: at θ = 0 and r = a, the left side is $1 + \ln 1 = 1 = \cos 0$ ✓

38 $xy^3\, dx + e^x\, dy = 0$; when $x \to \infty$, $y = \frac{1}{2}$ — Separable + IC

SOLUTION

Divide by $e^x y^3$:

$$xe^{-x}\,dx + y^{-3}\,dy = 0$$

Integrate (by parts: $\int xe^{-x}dx = -(x+1)e^{-x}$):

$$-(x+1)e^{-x} - \frac{1}{2y^2} = C$$

Apply $x \to \infty$: $(x+1)e^{-x} \to 0$ (the exponential wins), and $y = \frac{1}{2}$ gives $-\dfrac{1}{2(1/4)} = -2$, so $C = -2$.

$$\frac{1}{2y^2} = 2 - (x+1)e^{-x}$$

✓ ANSWER

$$\frac{1}{y^2} = 4 - 2(x+1)e^{-x} \qquad \text{or} \qquad y^2\left[4e^x - 2(x+1)\right] = e^x$$

IN SIMPLE WORDS "As x → ∞" is still just an initial condition. Take the limit of every x-term, then solve for C.

39 $\left(y - \sqrt{x^2+y^2}\right)dx - x\,dy = 0$; when $x = 0$, $y = 1$

Homogeneous + IC

SOLUTION

Every term is degree 1, so it is homogeneous. Let $y = vx$, $dy = v\,dx + x\,dv$ (take $x > 0$):

$$\left(vx - x\sqrt{1+v^2}\right)dx - x(v\,dx + x\,dv) = 0 \;\Rightarrow\; -x\sqrt{1+v^2}\,dx - x^2\,dv = 0$$

$$\frac{dv}{\sqrt{1+v^2}} = -\frac{dx}{x} \;\Rightarrow\; \ln\left(v + \sqrt{1+v^2}\right) = -\ln x + \ln C$$

So $v + \sqrt{1+v^2} = \dfrac{C}{x}$. Substitute $v = \frac{y}{x}$ and multiply by x:

$$y + \sqrt{x^2+y^2} = C$$

Apply $x = 0$, $y = 1$: $1 + 1 = C \Rightarrow C = 2$

$$\sqrt{x^2+y^2} = 2 - y \;\Rightarrow\; x^2 + y^2 = 4 - 4y + y^2$$

✓ ANSWER

$$x^2 = 4 - 4y \qquad \left(\text{a parabola: } y = 1 - \tfrac{x^2}{4}\right)$$

IN SIMPLE WORDS This is the particular solution tested in Long Test #13, which is why that item only works with the minus sign. Integral to remember: $\displaystyle\int \frac{dv}{\sqrt{1+v^2}} = \ln\left(v + \sqrt{1+v^2}\right)$.

40 $L\dfrac{di}{dt} + Ri = E$; when $t = 0$, $i = 0$ (L, R, E constants)

Linear + IC

SOLUTION

Standard form: $\dfrac{di}{dt} + \dfrac{R}{L}i = \dfrac{E}{L}$. IF $= e^{Rt/L}$.

$$\frac{d}{dt}\left(ie^{Rt/L}\right) = \frac{E}{L}e^{Rt/L} \;\Rightarrow\; ie^{Rt/L} = \frac{E}{R}e^{Rt/L} + C \;\Rightarrow\; i = \frac{E}{R} + Ce^{-Rt/L}$$

Apply $t = 0$, $i = 0$: $0 = \dfrac{E}{R} + C \Rightarrow C = -\dfrac{E}{R}$

✓ ANSWER

$$i(t) = \frac{E}{R}\left(1 - e^{-Rt/L}\right)$$

IN SIMPLE WORDS This is an RL circuit. The current starts at 0 and rises toward the steady value E/R (Ohm's law). L/R is the "time constant". (It is also separable: $\frac{L\,di}{E-Ri} = dt$.)

41 $3y(x^2-1)\,dx + (x^3+8y-3x)\,dy = 0$; when $x=0$, $y=1$ — Exact + IC

SOLUTION

$M = 3x^2y - 3y,\ \ N = x^3 + 8y - 3x$

Test: $M_y = 3x^2 - 3 = N_x$ ✓ Exact.

$$F = \int (3x^2y - 3y)\,dx = x^3y - 3xy + g(y)$$

$$F_y = x^3 - 3x + g'(y) = x^3 + 8y - 3x \ \Rightarrow\ g'(y) = 8y \ \Rightarrow\ g(y) = 4y^2$$

General solution: $x^3y - 3xy + 4y^2 = C$

Apply $x=0$, $y=1$: $0 - 0 + 4 = C \Rightarrow C = 4$

✓ ANSWER

$x^3y - 3xy + 4y^2 = 4$ or $xy(x^2-3) + 4y^2 = 4$

42 $(y^4 - 2xy)\,dx + 3x^2\,dy = 0$; when $x=2$, $y=1$ — Bernoulli + IC

SOLUTION

Solve for y': $y' = \dfrac{2xy - y^4}{3x^2}$, so

$$y' - \frac{2}{3x}\,y = -\frac{1}{3x^2}\,y^4 \qquad (n=4)$$

Let $v = y^{1-4} = y^{-3}$, $v' = -3y^{-4}y'$. Multiply the DE by $-3y^{-4}$:

$$v' + \frac{2}{x}\,v = \frac{1}{x^2}$$

IF $= x^2$:

$$(x^2v)' = 1 \ \Rightarrow\ x^2v = x + C \ \Rightarrow\ \frac{x^2}{y^3} = x + C$$

Apply $x=2$, $y=1$: $4 = 2 + C \Rightarrow C = 2$

✓ ANSWER

$x^2 = y^3(x+2)$ or $y^3 = \dfrac{x^2}{x+2}$

IN SIMPLE WORDS Check: differentiating $y^3 = \frac{x^2}{x+2}$ gives $3y^2y' = \frac{x^2+4x}{(x+2)^2}$, and the DE gives the same expression ✓

PART 3 Answer Key at a Glance

Long Test

1	PDE, order 2, degree 1
2	Order 1, degree 1; $y = x\sin^{-1}x + \sqrt{1-x^2} + C$
3	Order 3, degree 1
4	Order 2, degree 1
5	Order 2, degree 2
6	Order 2, degree 8
7	Linear
8	Linear
9	Nonlinear ($\sqrt{y'}$)
10	Nonlinear ($\cos(x+y)$)
11	Nonlinear (y in denominator)
12	Nonlinear (y^2, yy')
13	No (as printed); yes with "−"
14	Yes, is a solution
15	No (it solves $y'' - 2y' + y = 0$)
16	Yes, is a solution
17	Yes, is a solution
18	$y'' - 8y' + 16y = 0$
19	$\ddot{x} + \omega^2 x = 0$
20	$y'' - 6y' + 13y = 0$
21	$(x^2+x)y'' + (2x^2-1)y' - (4x+2)y = 0$
22	$(y - xy')^2 = Q^2[1 + (y')^2]$
23	$(x-r)^2[1+(y')^2] = r^2(y')^2$
24	$[1+(y')^2]^3 = (y'')^2$
25	$2a\,y'' + (y')^3 = 0$

Midterm Exam

1–14	See the comparison tables in Part 2-A
15	Order 3, degree 4
16	Order 2, degree 3
17	Order 2, degree 1
18	PDE, order 2, degree 1 (Laplace)
19	Verified: $u_{xx} + u_{yy} = 0$
20	Verified: $xy'' + y' + xy = 0$
21	$y'' - 4y' + 4y = 0$
22	$\ddot{x} + \omega^2 x = 0$
23	$y'' - 4y' + 13y = 0$
24	$(x^2-x)y'' - (2x^2-1)y' + (4x-2)y = 0$
25	$(y - xy')(y' - 1) = \alpha y'$
26	$[1+(y')^2]y''' - 3y'(y'')^2 = 0$
27	$3y''y^{(4)} - 5(y''')^2 = 0$
28	$2k\,y'' = 1$
29	$(x-a)y' = y - b$
30	$y'y''' - 3(y'')^2 = 0$
31	$\sin y = C\cos x$
32	$x\ln x - x + \ln(\ln y) = C$
33	$x\ln x + y\ln\frac{y}{x} - y = Cx$
34	$y = \frac{1}{4} - \frac{x}{2}\cot 2x + C\csc 2x$
35	$\sin x\cos y - \ln\lvert\sin x\rvert = C$
36	$y^2 = \dfrac{x^3}{C-x}$
37	$\frac{a^2}{r^2} + \ln\frac{r}{a} = \cos\theta$
38	$\frac{1}{y^2} = 4 - 2(x+1)e^{-x}$
39	$x^2 = 4 - 4y$
40	$i = \frac{E}{R}\left(1 - e^{-Rt/L}\right)$
41	$x^3y - 3xy + 4y^2 = 4$
42	$x^2 = y^3(x+2)$

Kaya mo 'yan, Sam! 💪 Solve each item on your own first, then compare with the solution.